

COMPLETE USER MANUAL

C++ Library Management System

1. System Overview :

The Library Management System is a C++ console application designed to handle the daily operations of a library. It uses Object-Oriented Programming (OOP) and stores all data permanently on the hard drive using binary files (book.dat and student.dat).

2. Starting the Application:

When the program starts, it clears the screen and displays a welcome banner. Press the Enter key to proceed.

Screen Output:

Plaintext

```
=====
||  LIBRARY MANAGEMENT SYSTEM  ||
||  (OOP Architecture)        ||
=====
```

Developed as a core C++ Project

Press Enter to continue...

3. The Main Menu (User Operations)

The Main Menu is for daily borrowing and returning.

Screen Output:

Plaintext

```
=====
MAIN MENU
=====

1. BOOK ISSUE
2. BOOK DEPOSIT
3. ADMINISTRATOR LOGIN
4. EXIT APPLICATION

Enter Your Choice (1-4):
```

Option 1: Book Issue

Allows a student to borrow a book. The system checks if the student exists and if they already have a book (students can only borrow 1 book at a time).

Screen Output:

Plaintext

```
--- BOOK ISSUE MODULE ---

Enter Student Admission No.: 101
Enter The Book No.: B01

[Success] Book Issued Successfully!
Note: Return the book within 15 days, otherwise a fine of Rs 15 per day applies.

Press Enter to continue...
```

Option 2: Book Deposit

Allows a student to return a book. The system asks how many days they kept it. If it is more than 15 days, a fine of Rs 15 per extra day is applied.

Screen Output:

Plaintext

--- BOOK DEPOSIT MODULE ---

Enter Admission No. Of Student: 101

Book Number : B01

Book Name : Data Structures

Book Author Name : Seymour Lipschutz

Book Deposited In No. Of Days: 17

[Penalty] Fine to be paid = Rs 30

[Success] Book Deposited Successfully.

Press Enter to continue...

4. Administrator Menu (System Management)

By choosing Option 3 from the Main Menu, you enter the Admin section. You must enter the security password (admin123).

Screen Output:

Plaintext

[SECURITY] Enter Administrator Password: admin123

=====

ADMINISTRATOR MENU

=====

1. CREATE STUDENT RECORD
2. DISPLAY ALL STUDENT RECORDS
3. DISPLAY SPECIFIC STUDENT
4. MODIFY STUDENT RECORD
5. DELETE STUDENT RECORD
6. CREATE BOOK RECORD
7. DISPLAY ALL BOOKS
8. DISPLAY SPECIFIC BOOK
9. MODIFY BOOK RECORD
10. DELETE BOOK RECORD
11. CLEAR ALL DATA (Factory Reset)
12. LOGOUT (Back To Main Menu)

Enter Your Choice (1-12):

Feature 1: Create Student Record

Registers a new student into the database.

Screen Output:

Plaintext

--- NEW STUDENT ENTRY ---

Enter The Admission No.: 102

Enter The Student Name: Alice Smith

[Success] Student Record Created.

Do you want to add another record? (y/n): n

Feature 2: Display All Student Records

Shows a formatted table of all students. "Book Issued" shows 0 if they have no book, and 1 if they have borrowed one.

Screen Output:

Plaintext

STUDENT LIST

```
=====
Admission No.   Name                Book Issued
=====
```

```
101             John Doe           1
102             Alice Smith         0
```

Press Enter to continue...

Feature 3: Display Specific Student

Searches for one student by Admission Number.

Screen Output:

Plaintext

Enter Admission No.: 101

--- STUDENT DETAILS ---

Admission Number : 101

Student Name : John Doe

No of Book Issued: 1

Issued Book No. : B01

Press Enter to continue...

Feature 4: Modify Student Record

Updates a student's name.

Screen Output:

Plaintext

--- MODIFY STUDENT RECORD ---

Enter the Admission No.: 102

Admission Number : 102

Student Name : Alice Smith

No of Book Issued: 0

--- Enter the new details ---

Admission No. 102

Modify Student Name: Alice Johnson

[Success] Record Updated

Press Enter to continue...

Feature 5: Delete Student Record

Permanently removes a student from the database.

Screen Output:

Plaintext

--- DELETE STUDENT ---

Enter the Admission No.: 102

[Success] Record Deleted.

Press Enter to continue...

Feature 6: Create Book Record

Adds a new book to the library catalog.

Screen Output:

Plaintext

--- NEW BOOK ENTRY ---

ENTER BOOK NO.: B02

ENTER BOOK NAME: Let Us C

ENTER AUTHOR NAME: Yashavant Kanetkar

[Success] Book Record Created.

Do you want to add another record? (y/n): n

Feature 7: Display All Books

Shows a formatted list of all books in the library.

Screen Output:

```
Plaintext
      BOOK LIST

=====
=
Book No.    Book Name        Author
=====
=
B01         Data Structures    Seymour Lipschutz
B02         Let Us C           Yashavant Kanetkar

Press Enter to continue...
```

Feature 8: Display Specific Book

Searches for a book by its Book Number.

Screen Output:

```
Plaintext
      Enter Book No.: B02

--- BOOK DETAILS ---

Book Number   : B02
Book Name     : Let Us C
Book Author Name : Yashavant Kanetkar

Press Enter to continue...
```

Feature 9: Modify Book Record

Changes the Title or Author of an existing book.

Screen Output:

Plaintext

--- MODIFY BOOK RECORD ---

Enter the Book No.: B02

Book Number : B02

Book Name : Let Us C

Book Author Name : Yashavant Kanetkar

--- Enter the new details ---

Book Number: B02

Modify Book Name: Let Us C++

Modify Author's Name: Yashavant Kanetkar

[Success] Record Updated

Press Enter to continue...

Feature 10: Delete Book Record

Permanently removes a book from the catalog.

Screen Output:

Plaintext

--- DELETE BOOK ---

Enter the Book No.: B02

[Success] Record Deleted.

Press Enter to continue...

Feature 11: Clear All Data (Factory Reset)

A critical feature that completely wipes the student.dat and book.dat files. Requires a y/n confirmation.

Screen Output:

Plaintext

--- WARNING: FACTORY RESET ---

Are you sure you want to delete ALL Students and Books data?

This cannot be undone! Type (y/n): y

[Success] All data has been permanently deleted. System is empty now.

Press Enter to continue...

Feature 12: Logout

Selecting option 12 exits the Administrator Menu and takes the user back to the Main Menu.

5. System Requirements & Background Info

- **Database System:** The system does not use SQL. It uses C++ File Handling (via the <fstream> library) to read and write binary files.
- **Error Handling:** If an incorrect ID is entered during a search, modification, or deletion, the system will output [Error] Record Not Found.
- **Security:** If an incorrect password is typed to enter the Admin menu, the system outputs [ERROR] Incorrect Password! Access Denied. and returns to the Main Menu.